LECTURE NOTES · ESS IN DAY-AHEAD MARKETS

Storage bids and Baldick's SOC-based design

The complete formulation: both market designs, their KKT conditions, the equivalence of the bids, and how the bids follow from technical data and prices

1 Notation

2 Current design

3 SOC-based design

4 Equivalence

5 Bids from the value of storage

6 Bids from prices

7 Non-uniqueness

8 Formal results

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What these notes prove, step by step

ROADMAP

1–3	**Formulation**	Write the current design (interval bids) and Baldick's SOC design, with all KKT conditions.
4	**Equivalence**	Bids built from the SOC design's multipliers reproduce its dispatch and prices (Prop. 1–3).
5	**Value of storage**	The bids only need the value of stored energy at the start and end of each interval (Cor. 3).
6	**Prices**	That value is read from nodal prices at marginal intervals (Lemmas 1–3, Prop. 4, Cor. 4).
7	**Non-uniqueness**	Where an SOC limit binds, prices only give a range; a selection rule is needed.
8	**Formal results**	No multipliers needed at all: bids = Φ(technical data, local prices) (Thm. 1, Prop. 5).

Key idea The central object is π, the value of one MWh of stored energy. Everything reduces to computing it from prices.

Sets, indices and variables

1 · NOTATION

Sets and indices		Variables	
$t \in \mathcal{T} = \{1, \ldots, T\}$	intervals of length ΔT (h)	$q_{i,k,t}$, $Q_{i,t}$	generation per block; unit output (MW)
$n, m \in \mathcal{N}$, r	buses; reference bus	$d_{m,j,t}$	served elastic demand (MW)
$(n, m) \in \Lambda$	lines, from bus n to bus m	$f_{n,m,t}$, $\theta_{n,t}$	flow $n \to m$ (MW); angle of bus n (rad)
$i \in \mathcal{G}$, $k \in \mathcal{K}_i$	units; offer blocks	$p_{e,t}$, $g_{e,t}$	ESS charge, discharge power (MW)
$m \in \mathcal{D}$, $j \in \mathcal{J}_m$	elastic loads; bid blocks	$s_{e,t}$	SOC at the *end* of interval t (MWh)
$e \in \mathcal{E}$	storage units (ESS)	$p_{e,j,t}$, $g_{e,k,t}$	ESS bid / offer blocks (current design)
$\mathcal{G}(n), \mathcal{D}(n), \mathcal{E}(n)$	units, loads, ESS at bus n		

Key idea Power is in MW over intervals of length ΔT; energy (SOC) is in MWh. Intervals are labelled by their end: s(e,t) is the SOC after interval t.

Data and multipliers

1 · NOTATION

ESS data		Multipliers ($/MWh)	
α_e	charge coefficient (1 MW for ΔT stores $\alpha_e \Delta T$ MWh)	$\lambda_{n,t}$	nodal price, balance (A1)
$\beta_e > \alpha_e$	inverse discharge coefficient	$\kappa_{n,m}, \xi, \bar{\mu}_{n,m}, \underline{\mu}_{n,m}$	network rows (A2)–(A4)
$\gamma_e \leq 1$	retention per interval	δ, τ, ψ	offer blocks, ramps, demand blocks
$\bar{s}_e, \underline{s}_e$; $\bar{p}_e, \bar{g}_e$	SOC limits (MWh); power limits (MW)	$\pi_{e,t}$	SOC dynamics (A8): **value of stored energy**
ϵ_e, ζ_e	degradation cost of charging / discharging	$\bar{\nu}_{e,t}, \underline{\nu}_{e,t}$	SOC limits (A9)–(A10): **scarcity rents**
$s_{e,0}$; $\bar{\bar{s}}_{e,T} = \gamma_e^T s_{e,0}$	initial SOC; final SOC if idle	$\bar{\rho}, \underline{\rho}, \bar{\sigma}, \underline{\sigma}$	charge / discharge power bounds
$b_e(\cdot)$, w_e	terminal bid on final SOC; its slope	$B_{n,m}$, $\bar{f}_{n,m}$	line susceptance; thermal limit (MW)
c^g, u^p; $\hat{g}, \hat{p}$	CD offer / bid prices; block sizes		

Key idea Rows in MW enter the Lagrangian with factor ΔT and rows in MWh without it, so every multiplier is in $/MWh.

Interval-by-interval bids, no SOC in the clearing

Objective: minimise the cost of the cleared offers minus the value of the cleared bids

$$(\mathcal{P}^{\mathrm{CD}}):\ \min \sum_{t\in\mathcal{T}} \Delta T\Big[\sum_{i,k} c_{i,k} q_{i,k,t} - \sum_{m,j} u_{m,j} d_{m,j,t} + \sum_{e}\Big(\sum_{k} c^{g}_{e,k,t}\, g_{e,k,t} - \sum_{j} u^{p}_{e,j,t}\, p_{e,j,t}\Big)\Big]$$

Constraints (multipliers in brackets)

$\sum_{m\in\mathcal{D}(n),j} d_{m,j,t} + D_{n,t} + \sum_{m:(n,m)\in\Lambda} f_{n,m,t} - \sum_{m:(m,n)\in\Lambda} f_{m,n,t}$ $- \sum_{i\in\mathcal{G}(n),k} q_{i,k,t} - \sum_{e\in\mathcal{E}(n)} (g_{e,t} - p_{e,t}) = 0$	$(\Delta T \lambda_{n,t})$	(A1) balance
$f_{n,m,t} - B_{n,m}(\theta_{n,t} - \theta_{m,t}) = 0, \quad (n,m)\in\Lambda$	$(\Delta T \kappa_{n,m,t})$	(A2) DC flow
$-\bar{f}_{n,m} \le f_{n,m,t} \le \bar{f}_{n,m}$	$(\Delta T \underline{\mu}_{n,m,t}, \Delta T \bar{\mu}_{n,m,t})$	(A3) line limits
$\theta_{r,t} = 0$	$(\Delta T \xi_t)$	(A4) reference
$0 \le q_{i,k,t} \le \bar{q}_{i,k,t}$	$(\Delta T \underline{\delta}, \Delta T \bar{\delta})$	(A5) offer blocks
$Q_{i,t} - Q_{i,t-1} \le \bar{R}_i, \quad Q_{i,t-1} - Q_{i,t} \le \underline{R}_i, \quad t \ge 2$	$(\Delta T \bar{\tau}, \Delta T \underline{\tau})$	(A6) ramps
$0 \le d_{m,j,t} \le \bar{d}_{m,j}$	$(\Delta T \underline{\psi}, \Delta T \bar{\psi})$	(A7) demand blocks
$0 \le g_{e,k,t} \le \hat{g}_{e,k,t}, \quad \sum_k \hat{g}_{e,k,t} \le \bar{g}_e$	$(\Delta T \underline{\sigma}, \Delta T \bar{\sigma})$	(C1) ESS offer blocks
$0 \le p_{e,j,t} \le \hat{p}_{e,j,t}, \quad \sum_j \hat{p}_{e,j,t} \le \bar{p}_e$	$(\Delta T \underline{\rho}, \Delta T \bar{\rho})$	(C2) ESS bid blocks

Key idea The ESS appears as a generator (offers c^g) plus a load (bids u^p) in every interval. Nothing links interval t to t+1: the SOC is the owner's problem.

KKT conditions of the current design

2 · CURRENT DESIGN (CD)

Stationarity

$$
\begin{aligned}
\partial q_{i,k,t}: &\quad c_{i,k} - \lambda_{n(i),t} + \bar{\delta}_{i,k,t} - \underline{\delta}_{i,k,t} + \varrho_{i,t} = 0 && \text{(S1)}\\
\partial d_{m,j,t}: &\quad -u_{m,j} + \lambda_{n(m),t} + \bar{\psi}_{m,j,t} - \underline{\psi}_{m,j,t} = 0 && \text{(S2)}\\
\partial f_{n,m,t}: &\quad \lambda_{n,t} - \lambda_{m,t} + \kappa_{n,m,t} + \bar{\mu}_{n,m,t} - \underline{\mu}_{n,m,t} = 0 && \text{(S3)}\\
\partial \theta_{n,t}: &\quad -\textstyle\sum_{m:(n,m)\in\Lambda} B_{n,m}\kappa_{n,m,t} + \sum_{m:(m,n)\in\Lambda} B_{m,n}\kappa_{m,n,t} + \xi_t\,\mathbf{1}_{\{n=r\}} = 0 && \text{(S4)}\\
\partial g_{e,k,t}: &\quad c^g_{e,k,t} - \lambda_{n(e),t} + \bar{\sigma}_{e,k,t} - \underline{\sigma}_{e,k,t} = 0 && \text{(S5}^{\text{CD}}\text{)}\\
\partial p_{e,j,t}: &\quad -u^p_{e,j,t} + \lambda_{n(e),t} + \bar{\rho}_{e,j,t} - \underline{\rho}_{e,j,t} = 0 && \text{(S6}^{\text{CD}}\text{)}
\end{aligned}
$$

$$\varrho_{i,t} := (\bar{\tau}_{i,t} - \underline{\tau}_{i,t}) - (\bar{\tau}_{i,t+1} - \underline{\tau}_{i,t+1}) \qquad \text{(ramp wedge)}$$

Box form of the ESS rows

$$0 \le g_{e,k,t} \le \hat{g}_{e,k,t} \;\perp\; c^g_{e,k,t} - \lambda_{n(e),t}, \qquad 0 \le p_{e,j,t} \le \hat{p}_{e,j,t} \;\perp\; \lambda_{n(e),t} - u^p_{e,j,t}$$

In words

- (S1)–(S4): the rest of the market; identical in both designs.
- (S5-CD): an offer block clears fully if c^g < λ, not at all if c^g > λ, partially only if c^g = λ.
- (S6-CD): a bid block clears fully if u^p > λ, not at all if u^p < λ.
- Complementary slackness (CS1)–(CS4-CD) pairs each bound with its multiplier.

Key idea In the CD an ESS is just a set of price-quantity steps. Its intertemporal plan must be built into those prices.

The ISO optimises the state of charge

Objective: energy cost + ESS degradation − value of the final SOC

$$(\mathcal{P}^{\mathrm{SOC}}):\ \min \sum_t \Delta T\Big[\sum_{i,k} c_{i,k}q_{i,k,t} - \sum_{m,j} u_{m,j}d_{m,j,t} + \sum_e (\epsilon_e p_{e,t} + \zeta_e g_{e,t})\Big] - \sum_e b_e\big(s_{e,T} - \bar{\bar{s}}_{e,T}\big)$$

Constraints

(A1)–(A7)		rest of market, as in CD
$s_{e,t} - \gamma_e s_{e,t-1} - \Delta T\alpha_e p_{e,t} + \Delta T\beta_e g_{e,t} = 0$	$(\pi_{e,t})$	(A8) SOC dynamics
$\gamma_e s_{e,t-1} + \Delta T\alpha_e p_{e,t} - \bar{s}_e \le 0$	$(\bar{\nu}_{e,t})$	(A9) cannot overfill
$\underline{s}_e - \gamma_e s_{e,t-1} + \Delta T\beta_e g_{e,t} \le 0$	$(\underline{\nu}_{e,t})$	(A10) cannot overdrain
$0 \le p_{e,t} \le \bar{p}_e$	$(\Delta T\underline{\rho}_{e,t}, \Delta T\bar{\rho}_{e,t})$	(A11) charge power
$0 \le g_{e,t} \le \bar{g}_e$	$(\Delta T\underline{\sigma}_{e,t}, \Delta T\bar{\sigma}_{e,t})$	(A12) discharge power

In words

- The ESS submits only θ (technical data) and b_e.
- (A9)–(A10) are Baldick's tightened limits: they use the SOC at the start of t and the power of interval t.
- $\bar{\bar{s}}$ is the final SOC if the ESS stayed idle.

Key idea The ISO now sees the SOC dynamics, so it can schedule the ESS optimally without any price forecast by the owner.

Lagrangian and KKT conditions of the ESS

Lagrangian (ESS terms shown; scaling makes every multiplier $/MWh)

$$\mathcal{L}^{\mathrm{SOC}} = f^{\mathrm{SOC}} + \sum_{n,t} \Delta T \lambda_{n,t}[\mathrm{A1}]_{n,t} + \cdots \ \text{(rest of market, all with } \Delta T\text{)}$$
$$+ \sum_{e,t} \left\{ \pi_{e,t}[\mathrm{A8}]_{e,t} + \bar{\nu}_{e,t}[\mathrm{A9}]_{e,t} + \underline{\nu}_{e,t}[\mathrm{A10}]_{e,t} \right\} \ \text{(energy rows: no } \Delta T\text{)}$$
$$+ \sum_{e,t} \Delta T \left[\bar{\rho}_{e,t}(p_{e,t} - \bar{p}_e) - \underline{\rho}_{e,t} p_{e,t} + \bar{\sigma}_{e,t}(g_{e,t} - \bar{g}_e) - \underline{\sigma}_{e,t} g_{e,t} \right]$$

Stationarity of the ESS variables

$$\partial p_{e,t}: \quad \epsilon_e + \lambda_{n(e),t} - \alpha_e(\pi_{e,t} - \bar{\nu}_{e,t}) + \bar{\rho}_{e,t} - \underline{\rho}_{e,t} = 0 \quad \text{(S5)}$$
$$\partial g_{e,t}: \quad \zeta_e - \lambda_{n(e),t} + \beta_e(\pi_{e,t} + \underline{\nu}_{e,t}) + \bar{\sigma}_{e,t} - \underline{\sigma}_{e,t} = 0 \quad \text{(S6)}$$
$$\partial s_{e,t},\ t < T: \quad \pi_{e,t} - \gamma_e(\pi_{e,t+1} - \bar{\nu}_{e,t+1} + \underline{\nu}_{e,t+1}) = 0 \quad \text{(S7)}$$
$$\partial s_{e,T}: \quad \pi_{e,T} \in \partial b_e(s_{e,T} - \bar{\bar{s}}_{e,T}) \quad (= w_e \text{ for linear } b_e) \quad \text{(S8)}$$

In words

- (S5)–(S6): the price equals the ESS's marginal value of charging / discharging.
- (S7): how the value of stored energy moves from t to t+1.
- (S8): at the end, the value equals the terminal bid's slope.

Key idea The rest-of-market rows (S1)–(S4) are the same as in the CD. Only the ESS rows differ.

Complementary slackness and the value of stored energy

Complementary slackness of the ESS

$$0 \le \underline{\rho}_{e,t} \perp p_{e,t}, \quad 0 \le \bar{\rho}_{e,t} \perp \bar{p}_e - p_{e,t} \ge 0, \quad 0 \le \underline{\sigma}_{e,t} \perp g_{e,t}, \quad 0 \le \bar{\sigma}_{e,t} \perp \bar{g}_e - g_{e,t} \ge 0 \quad \text{(CS4)}$$

$$0 \le \bar{\nu}_{e,t} \perp \bar{s}_e - \gamma_e s_{e,t-1} - \Delta T \alpha_e p_{e,t} \ge 0, \quad 0 \le \underline{\nu}_{e,t} \perp \gamma_e s_{e,t-1} - \Delta T \beta_e g_{e,t} - \underline{s}_e \ge 0 \quad \text{(CS5)}$$

Box form

$$0 \le p_{e,t} \le \bar{p}_e \perp \lambda_{n(e),t} + \epsilon_e - \alpha_e(\pi_{e,t} - \bar{\nu}_{e,t}), \quad 0 \le g_{e,t} \le \bar{g}_e \perp \zeta_e + \beta_e(\pi_{e,t} + \underline{\nu}_{e,t}) - \lambda_{n(e),t}$$

Unrolling (S7) backwards from (S8)

$$\pi_{e,t} = \underbrace{\gamma_e^{T-t}\pi_{e,T}}_{\text{terminal value}} + \underbrace{\sum_{\tau=t+1}^{T} \gamma_e^{\tau-t}\left(\underline{\nu}_{e,\tau} - \bar{\nu}_{e,\tau}\right)}_{\text{future SOC rents}}$$

π(e,t) = value of 1 MWh stored at the end of t

- = discounted terminal value
- + future SOC rents: positive when the ESS will be empty ($\underline{\nu}$), negative when full ($\bar{\nu}$).
- Without binding limits, π = γ^(T−t)·w.

Key idea π is the economic heart of the problem: every bid below is a function of π.

Before the equivalence: the KKT rows we compare

Formulas we use (with their labels)

Current design

$$(\mathrm{S5^{CD}}) \quad c^{g}_{e,k,t} - \lambda_{n(e),t} + \bar{\sigma}_{e,k,t} - \underline{\sigma}_{e,k,t} = 0 \qquad \text{offer block}$$

$$(\mathrm{S6^{CD}}) \quad -u^{p}_{e,j,t} + \lambda_{n(e),t} + \bar{\rho}_{e,j,t} - \underline{\rho}_{e,j,t} = 0 \qquad \text{bid block}$$

SOC design

$$(\mathrm{S5}) \quad \epsilon_e + \lambda_{n(e),t} - \alpha_e(\pi_{e,t} - \bar{\nu}_{e,t}) + \bar{\rho}_{e,t} - \underline{\rho}_{e,t} = 0 \qquad \text{charge}$$

$$(\mathrm{S6}) \quad \zeta_e - \lambda_{n(e),t} + \beta_e(\pi_{e,t} + \underline{\nu}_{e,t}) + \bar{\sigma}_{e,t} - \underline{\sigma}_{e,t} = 0 \qquad \text{discharge}$$

$$(\mathrm{S7}) \quad \pi_{e,t} = \gamma_e(\pi_{e,t+1} - \bar{\nu}_{e,t+1} + \underline{\nu}_{e,t+1}), \quad t < T \qquad \text{value over time}$$

$$(\mathrm{S8}) \quad \pi_{e,T} = w_e \qquad \text{terminal value}$$

$$(\mathrm{CS4}) \quad 0 \le \underline{\rho}_{e,t} \perp p_{e,t} \ge 0, \quad 0 \le \bar{\rho}_{e,t} \perp \bar{p}_e - p_{e,t} \ge 0 \quad (\text{same for } \sigma, g) \qquad \text{power bounds}$$

Notation

$\lambda_{n,t}$	nodal price at bus n, interval t (\$/MWh); balance (A1)
$u^p,\ c^g$	CD bid price (to charge) and offer price (to discharge)
$\hat{p},\ \hat{g}$	CD block sizes (MW)
$\bar{p}_e,\ \bar{g}_e$	physical power limits (MW)
$\pi_{e,t}$	value of 1 MWh stored at the end of t; (A8)
$\bar{\nu}_{e,t},\ \underline{\nu}_{e,t}$	rents of the full / empty SOC limit; (A9)–(A10)
$\bar{\rho}, \underline{\rho}$ / $\bar{\sigma}, \underline{\sigma}$	multipliers of the charge / discharge power bounds
$\alpha_e,\ \beta_e,\ \gamma_e$	charge, inverse discharge, retention coefficients
$\epsilon_e,\ \zeta_e;\ w_e$	degradation costs; slope of the terminal bid

Used for Section 4 matches (S5)–(S6) of the SOC design with (S6-CD)–(S5-CD) of the current design, row by row.

KKT side by side: only the ESS rows differ

Condition	Current design	SOC design
Rest of market	(S1)–(S4), (CS1)–(CS3)	identical
Charge	$\lambda_{n(e),t} = u^{p}_{e,j,t} - \bar{\rho}_{e,j,t} + \underline{\rho}_{e,j,t}$	$\lambda_{n(e),t} = \left[\alpha_e(\pi_{e,t} - \bar{\nu}_{e,t}) - \epsilon_e\right] - \bar{\rho}_{e,t} + \underline{\rho}_{e,t}$
Discharge	$\lambda_{n(e),t} = c^{g}_{e,k,t} + \bar{\sigma}_{e,k,t} - \underline{\sigma}_{e,k,t}$	$\lambda_{n(e),t} = \left[\zeta_e + \beta_e(\pi_{e,t} + \underline{\nu}_{e,t})\right] + \bar{\sigma}_{e,t} - \underline{\sigma}_{e,t}$
Power bounds	$\hat{p}_{e,j,t}$, $\hat{g}_{e,k,t}$ (bid sizes)	$\bar{p}_e$, $\bar{g}_e$ (physical)
SOC rows	absent	(A8)–(A10), (S7)–(S8), (CS5)

Key idea Read the brackets: in the SOC design, the term in [·] plays exactly the role of the CD bid price. Matching them is the whole trick.

Equivalent bids and Proposition 1

Equivalent bids (one block per interval, block sizes = physical limits)

$$u^{p\star}_{e,t} = \alpha_e\left(\pi^\star_{e,t} - \bar{\nu}^\star_{e,t}\right) - \epsilon_e, \quad \hat{p}_{e,t} = \bar{p}_e; \qquad c^{g\star}_{e,t} = \zeta_e + \beta_e\left(\pi^\star_{e,t} + \underline{\nu}^\star_{e,t}\right), \quad \hat{g}_{e,t} = \bar{g}_e \qquad \text{(E)}$$

Proposition 1 (KKT equivalence)

Let z★ be primal–dual optimal for P_SOC. Under bids (E), the SOC-design dispatch with multipliers (λ★, ρ★, σ★, …) satisfies every KKT row of P_CD. Hence it is CD-optimal, with the same nodal prices λ★.

Proof

$$\text{(S5)}: \ \lambda_{n(e),t} = \underbrace{\alpha_e(\pi^\star_{e,t} - \bar{\nu}^\star_{e,t}) - \epsilon_e}_{=u^{p\star}_{e,t}} - \bar{\rho}^\star_{e,t} + \underline{\rho}^\star_{e,t} \implies \text{(S6}^{\text{CD}}\text{)}, \qquad \text{(S6)}: \ \lambda_{n(e),t} = \underbrace{\zeta_e + \beta_e(\pi^\star_{e,t} + \underline{\nu}^\star_{e,t})}_{=c^{g\star}_{e,t}} + \bar{\sigma}^\star_{e,t} - \underline{\sigma}^\star_{e,t} \implies \text{(S5}^{\text{CD}}\text{)}$$

Block sizes $\hat{p} = \bar{p}$ and $\hat{g} = \bar{g}$ make (CS4) identical to (CS4-CD); all other rows are common; P_CD is an LP, so KKT is sufficient.

Key idea The bids are the SOC design's dual prices: the value of the energy the ESS stores or gives up, net of degradation.

The CD as a relaxation, and what any bid must satisfy

Proposition 2 (CD = Lagrangian relaxation of the SOC rows)

$$\min_{s}\left\{f^{\mathrm{SOC}}+\sum_{e,t}\left(\pi^{\star}_{e,t}[\mathrm{A8}]_{e,t}+\bar{\nu}^{\star}_{e,t}[\mathrm{A9}]_{e,t}+\underline{\nu}^{\star}_{e,t}[\mathrm{A10}]_{e,t}\right)\right\}=f^{\mathrm{CD}}+K,\qquad K=\sum_{e}K_{e},$$

$$K_{e}=-\pi^{\star}_{e,0}s_{e,0}-\bar{s}_{e}\sum_{t}\bar{\nu}^{\star}_{e,t}+\underline{s}_{e}\sum_{t}\underline{\nu}^{\star}_{e,t}+\min_{s}\left\{\pi^{\star}_{e,T}s-b_{e}(s-\bar{\bar{s}}_{e,T})\right\},\quad \pi^{\star}_{e,0}:=\gamma_{e}(\pi^{\star}_{e,1}-\bar{\nu}^{\star}_{e,1}+\underline{\nu}^{\star}_{e,1})$$

Consequences: (i) f^SOC★ = f^CD★ + K; (ii) a CD optimum is SOC-optimal iff some SOC path satisfies (A8)–(A10) with complementary slackness.

Proposition 3 (necessary and sufficient, any stepped bids)

$$\sum_{j:\,u^{p}_{e,j,t}>\lambda^{\star}}\hat{p}_{e,j,t}\;\le\;p^{\star}_{e,t}\;\le\sum_{j:\,u^{p}_{e,j,t}\ge\lambda^{\star}}\hat{p}_{e,j,t},\qquad \sum_{k:\,c^{g}_{e,k,t}<\lambda^{\star}}\hat{g}_{e,k,t}\;\le\;g^{\star}_{e,t}\;\le\sum_{k:\,c^{g}_{e,k,t}\le\lambda^{\star}}\hat{g}_{e,k,t}$$

The SOC-design schedule must lie on the submitted bid and offer curves, evaluated at λ★. With bids (E): c^g − u^p > 0, so the ESS never charges and discharges at once.

$$c^{g\star}_{e,t}-u^{p\star}_{e,t}=\epsilon_{e}+\zeta_{e}+(\beta_{e}-\alpha_{e})\pi^{\star}_{e,t}+\beta_{e}\underline{\nu}^{\star}_{e,t}+\alpha_{e}\bar{\nu}^{\star}_{e,t}>0\implies p_{e,t}\,g_{e,t}=0$$

Key idea Prices are reproduced exactly; quantities can tie at the bid edges, so the CD may have other optima (a tie-break rule is needed).

What information the bids embed

4 · EQUIVALENCE

Corollary 1 (unrolled bids)

$$c^{g\star}_{e,t} = \zeta_e + \beta_e \left[\gamma_e^{T-t} \pi^{\star}_{e,T} + \sum_{\tau=t+1}^{T} \gamma_e^{\tau-t} \left(\underline{\nu}^{\star}_{e,\tau} - \bar{\nu}^{\star}_{e,\tau} \right) + \underline{\nu}^{\star}_{e,t} \right]$$

$$u^{p\star}_{e,t} = \alpha_e \left[\gamma_e^{T-t} \pi^{\star}_{e,T} + \sum_{\tau=t+1}^{T} \gamma_e^{\tau-t} \left(\underline{\nu}^{\star}_{e,\tau} - \bar{\nu}^{\star}_{e,\tau} \right) - \bar{\nu}^{\star}_{e,t} \right] - \epsilon_e$$

Two kinds of terms

- Private: α, β, γ, ε, ζ, w (exactly what the ESS submits in the SOC design).
- Endogenous: own SOC rents v★, which depend on the market only through λ★ at the ESS's node.

Corollary 2 (when private information is enough)

$$b_e(\Delta) = w_e \Delta, \quad \bar{\nu}^{\star}_{e,\tau} = \underline{\nu}^{\star}_{e,\tau} = 0 \ \forall \tau \geq t \implies c^{g\star}_{e,t} = \zeta_e + \beta_e \gamma_e^{T-t} w_e, \quad u^{p\star}_{e,t} = \alpha_e \gamma_e^{T-t} w_e - \epsilon_e$$

Otherwise the bids need the rents, i.e. a forecast of the ESS's own nodal prices (Baldick's critique of interval bids).

Key idea If no SOC limit ever binds, private data suffice, however volatile prices are. Information only matters at the limits.

Before Section 5: bids, value dynamics and SOC limits

Formulas we use (with their labels)

(E)	$u^p_{e,t} = \alpha_e(\pi_{e,t} - \bar{\nu}_{e,t}) - \epsilon_e, \quad c^g_{e,t} = \zeta_e + \beta_e(\pi_{e,t} + \underline{\nu}_{e,t})$	equivalent bids
(S7)	$\pi_{e,t-1} = \gamma_e(\pi_{e,t} - \bar{\nu}_{e,t} + \underline{\nu}_{e,t})$	written for $(t-1, t)$
(S8)	$\pi_{e,T} \in \partial b_e(s_{e,T} - \bar{\bar{s}}_{e,T}) \quad (= w_e)$	terminal value
(A9)	$\gamma_e s_{e,t-1} + \Delta T \alpha_e p_{e,t} \le \bar{s}_e \quad [\bar{\nu}_{e,t}]$	cannot overfill
(A10)	$\gamma_e s_{e,t-1} - \Delta T \beta_e g_{e,t} \ge \underline{s}_e \quad [\underline{\nu}_{e,t}]$	cannot overdrain
(CS5)	$0 \le \bar{\nu}_{e,t} \perp$ slack of (A9) $\ge 0, \quad 0 \le \underline{\nu}_{e,t} \perp$ slack of (A10) ≥ 0	rent only if binding

Notation

$\pi_{e,t}$	value of stored energy at the *end* of t
$\pi_{e,t-1}/\gamma_e$	the same value seen at the *start* of t
$s_{e,t-1}$	SOC at the start of t (end of $t-1$), MWh
$\bar{s}_e$, $\underline{s}_e$	maximum / minimum SOC (MWh)
$\bar{\nu}_{e,t}$ / $\underline{\nu}_{e,t}$	rent paid when the ESS is full / empty in t
b_e, w_e; $\bar{\bar{s}}_{e,T}$	terminal bid and its slope; final SOC if idle
ΔT	interval length (h); powers in MW, energy in MWh
$r_{e,t}$	**new:** net SOC rent $\underline{\nu}_{e,t} - \bar{\nu}_{e,t}$

Used for Corollary 3 removes $\bar{\nu}$ and $\underline{\nu}$ from (E) using (S7) for the pair (t−1, t), the limits (A9)–(A10) and (CS5).

Terminal value, net rent and Assumption 1

Terminal value, from (S8)

$$\pi^{\star}_{e,T} = w_e\left(s^{\star}_{e,T} - \bar{\bar{s}}_{e,T}\right) \quad (= w_e \text{ for linear } b_e) \qquad (11)$$

Net SOC rent: the jump of π across interval t, from (S7)

$$r^{\star}_{e,t} := \underline{\nu}^{\star}_{e,t} - \bar{\nu}^{\star}_{e,t} = \frac{\pi^{\star}_{e,t-1}}{\gamma_e} - \pi^{\star}_{e,t} \qquad (12)$$

Under Assumption 1, its sign identifies the active limit

$$\underline{\nu}^{\star}_{e,t} = \left[r^{\star}_{e,t}\right]^{+} \text{ (ESS empty)}, \qquad \bar{\nu}^{\star}_{e,t} = \left[-r^{\star}_{e,t}\right]^{+} \text{ (ESS full)} \qquad (13)$$

Assumption 1: at most one SOC limit is active per interval

$$\text{both (A9), (A10) tight} \implies \Delta T(\alpha_e p_{e,t} + \beta_e g_{e,t}) = \bar{s}_e - \underline{s}_e; \qquad \text{sufficient: } \Delta T(\alpha_e \bar{p}_e + \beta_e \bar{g}_e) < \bar{s}_e - \underline{s}_e$$

It can also hold without the sufficient condition (e.g. Baldick's base case, where the solver never returns two positive SOC multipliers in one interval).

Key idea Only the difference $\underline{\nu} - \bar{\nu}$ enters the dynamics. Assumption 1 lets us recover each multiplier from that difference.

Corollary 3: bids depend only on π at the start and end of t

$$c^{g\star}_{e,t} = \zeta_e + \beta_e \max\left\{\pi^\star_{e,t},\ \frac{\pi^\star_{e,t-1}}{\gamma_e}\right\}, \qquad u^{p\star}_{e,t} = \alpha_e \min\left\{\pi^\star_{e,t},\ \frac{\pi^\star_{e,t-1}}{\gamma_e}\right\} - \epsilon_e \tag{14}$$

Proof (steps 1–4)

1. $c^g_t = \zeta + \beta(\pi_t + \underline{\nu}_t), \qquad u^p_t = \alpha(\pi_t - \bar{\nu}_t) - \epsilon$
2. $\pi_{t-1} = \gamma(\pi_t - \bar{\nu}_t + \underline{\nu}_t), \quad 0 \le \bar{\nu}_t \perp \bar{s} - \gamma s_{t-1} - \Delta T \alpha p_t \ge 0, \quad 0 \le \underline{\nu}_t \perp \gamma s_{t-1} - \Delta T \beta g_t - \underline{s} \ge 0$
3. $r_t = \underline{\nu}_t - \bar{\nu}_t = \dfrac{\pi_{t-1}}{\gamma} - \pi_t$
4. $\underline{\nu}_t > 0,\ \bar{\nu}_t > 0 \ \Rightarrow\ \Delta T(\alpha p_t + \beta g_t) = \bar{s} - \underline{s}$ (excluded by Assumption 1) $\Rightarrow\ \underline{\nu}_t \bar{\nu}_t = 0$

Key idea Step 3 is (S7) written for the pair (t−1, t): it links the rent of interval t to the values at its start and end.

Corollary 3: proof (steps 5–8)

5 · BIDS FROM THE VALUE OF STORAGE

5.

$$\begin{aligned} &\underline{\nu}_t > 0: \quad \bar{\nu}_t = 0, \quad r_t = \underline{\nu}_t > 0 \quad \text{(empties)} \\ &\bar{\nu}_t > 0: \quad \underline{\nu}_t = 0, \quad r_t = -\bar{\nu}_t < 0 \quad \text{(fills)} \\ &\text{both } 0: \qquad\quad r_t = 0 \end{aligned}$$

6. $\underline{\nu}_t = \max\{0, r_t\}, \qquad \bar{\nu}_t = -\min\{0, r_t\} \qquad$ (check each case of step 5)

7.

$$\pi_t + \underline{\nu}_t \overset{(a)}{=} \pi_t + \max\{0, r_t\} \overset{(b)}{=} \pi_t + \max\left\{0, \tfrac{\pi_{t-1}}{\gamma} - \pi_t\right\} \overset{(c)}{=} \max\left\{\pi_t, \tfrac{\pi_{t-1}}{\gamma}\right\}$$

$$\pi_t - \bar{\nu}_t \overset{(a)}{=} \pi_t + \min\{0, r_t\} \overset{(b)}{=} \pi_t + \min\left\{0, \tfrac{\pi_{t-1}}{\gamma} - \pi_t\right\} \overset{(c)}{=} \min\left\{\pi_t, \tfrac{\pi_{t-1}}{\gamma}\right\}$$

Property used in step 7(c)

$$a + \max\{x, y\} = \max\{a + x, a + y\}, \qquad a + \min\{x, y\} = \min\{a + x, a + y\}$$

(a) v → r by step 6; (b) r → π by step 3; (c) move π_t inside the bracket. Step 8: substitute both brackets into step 1, which gives (14).

Check (base case, t = 2): $\pi_1 = 49$, $\pi_2 = 1.11 \Rightarrow r_2 = 47.89 = \underline{v}_2$, $\bar{v}_2 = 0 \Rightarrow c_2 = 1 + \max\{1.11, 49\} = 50$, $u_2 = 0.9 \cdot \min\{1.11, 49\} - 1 = 0$.



Key idea Offer uses the higher of the two values, bid the lower one: the bid–offer band opens exactly when the SOC hits a limit.

Before Section 6: bids in terms of π, and how a price reads π

Formulas we use (with their labels)

(14)	$c^g_{e,t} = \zeta_e + \beta_e \max\{\pi_{e,t}, \pi_{e,t-1}/\gamma_e\}, \quad u^p_{e,t} = \alpha_e \min\{\pi_{e,t}, \pi_{e,t-1}/\gamma_e\} - \epsilon_e$	Cor. 3
(12)	$r_{e,t} = \underline{\nu}_{e,t} - \bar{\nu}_{e,t} = \pi_{e,t-1}/\gamma_e - \pi_{e,t}$	net rent
(13)	$\underline{\nu}_{e,t} = [r_{e,t}]^+, \quad \bar{\nu}_{e,t} = [-r_{e,t}]^+$	Assumption 1
(S5)	$\epsilon_e + \lambda_{n(e),t} - \alpha_e(\pi_{e,t} - \bar{\nu}_{e,t}) + \bar{\rho}_{e,t} - \underline{\rho}_{e,t} = 0$	charge
(S6)	$\zeta_e - \lambda_{n(e),t} + \beta_e(\pi_{e,t} + \underline{\nu}_{e,t}) + \bar{\sigma}_{e,t} - \underline{\sigma}_{e,t} = 0$	discharge
(CS4)	$0 < p_{e,t} < \bar{p}_e \Rightarrow \bar{\rho}_{e,t} = \underline{\rho}_{e,t} = 0; \quad 0 < g_{e,t} < \bar{g}_e \Rightarrow \bar{\sigma}_{e,t} = \underline{\sigma}_{e,t} = 0$	partial trade

Notation

$\lambda_{n(e),t}$	nodal price at the ESS's bus $n(e)$
$p^\star_{e,t},\ g^\star_{e,t}$	SOC-design schedule (charge, discharge), MW
$r_{e,t}$	net rent: >0 ESS empties, <0 fills, $=0$ free
$[x]^+$	$\max\{0, x\}$
$\bar{\rho}, \underline{\rho}\ /\ \bar{\sigma}, \underline{\sigma}$	power-bound multipliers; zero when trade is partial
$\mathcal{H}_e$	**new:** binding intervals: $\underline{\nu}_{e,t} > 0$ or $\bar{\nu}_{e,t} > 0$
$\mathcal{S}_{e,k}$	**new:** segment k: from one binding interval to the next

Used for Lemma 1 reads π from (S5)–(S6) where (CS4) sets the power multipliers to zero; segments start where the rent r is non-zero.

Segments: where π can jump

$$\mathcal{H}_e = \{h_{e,1} < \cdots < h_{e,N_e}\}, \qquad \mathcal{S}_{e,k} = \{h_{e,k-1}, \ldots, h_{e,k} - 1\}, \quad h_{e,0} := 1; \qquad \mathcal{S}_{e,N_e+1} = \{h_{e,N_e}, \ldots, T\}$$

$$t, t+1 \in \mathcal{S}_{e,k} \implies r^\star_{e,t+1} = 0 \implies \pi^\star_{e,t} = \gamma_e \, \pi^\star_{e,t+1}$$

Base case, solver duals	t = 1	t = 2	t = 3	t = 4	t = 5	t = 6
ESS action (MW)	charge 100	discharge 90	charge 200	charge 22.2	discharge 175	discharge 25
SOC at end of t (MWh)	360	0	720	800	100	0
limit binding in t	–	**empty**	–	**full**	–	**empty**
rent r_t	0	47.89	0	−47.89	0	9
π_t	**49**	**1.11**	**1.11**	**49**	**49**	**40**
segment	S_1	S_2	S_2	S_3	S_3	S_4 (last)

- Between limits the ESS can shift energy freely, so one MWh is worth the same in every interval (up to retention).
- A binding interval h opens a new segment: (A9)–(A10) at h use the SOC at the end of h−1, so the rent sits between the value at the end of h−1 (old segment) and at the end of h (new segment).

Key idea Each segment carries one number. N+1 values describe the whole vector π instead of T.

Marginal intervals and Lemma 1: the price reveals the value

Definition (marginal interval of segment k)

$$\begin{aligned}&\text{(D) discharge-marginal:} \quad 0 < g^\star_{e,\tilde{t}} < \bar{g}_e \ \text{ and } \ \underline{\nu}^\star_{e,\tilde{t}} = 0\\ &\text{(C) charge-marginal:} \quad\ 0 < p^\star_{e,\tilde{t}} < \bar{p}_e \ \text{ and } \ \bar{\nu}^\star_{e,\tilde{t}} = 0\end{aligned}$$

Lemma 1

$$v_{e,k}(\lambda) = \begin{cases} \dfrac{\lambda_{n(e),\tilde{t}} - \zeta_e}{\beta_e}, & \text{(D)} \\ \dfrac{\lambda_{n(e),\tilde{t}} + \epsilon_e}{\alpha_e}, & \text{(C)} \end{cases} \quad \Longrightarrow \quad \pi^\star_{e,\tilde{t}} = v_{e,k}(\lambda^\star)$$

Proof

$$\text{(D)}: \ 0 < g < \bar{g} \overset{\text{(CS4)}}{\Rightarrow} \bar{\sigma} = \underline{\sigma} = 0, \ \underline{\nu} = 0 \overset{\text{(S6)}}{\Rightarrow} \zeta - \lambda + \beta\pi = 0 \ \Rightarrow \ \pi = \frac{\lambda - \zeta}{\beta}$$

$$\text{(C)}: \ 0 < p < \bar{p} \overset{\text{(CS4)}}{\Rightarrow} \bar{\rho} = \underline{\rho} = 0, \ \bar{\nu} = 0 \overset{\text{(S5)}}{\Rightarrow} \epsilon + \lambda - \alpha\pi = 0 \ \Rightarrow \ \pi = \frac{\lambda + \epsilon}{\alpha}$$

Interpretation: discharging 1 stored MWh earns (λ − ζ)/β; storing 1 MWh costs (λ + ε)/α. At a marginal interval the ESS is indifferent.

Why both conditions

- At full power the ESS would trade more: the price only bounds π.
- If it empties (fills) in that interval, the limit stops it, not the price.
- A binding interval can still be marginal for the other action.

Key idea Where the ESS trades partially and freely, the market price is exactly its value of storage.

Lemmas 2–3: carrying the value, and which price sets it

Lemma 2 (the value is carried through the segment)

$$t < t' \in \mathcal{S}_{e,k}: \quad \pi^\star_{e,j-1} = \gamma_e \pi^\star_{e,j} \ \ (j = t+1, \dots, t') \implies \pi^\star_{e,t} = \gamma_e^{t'-t}\, \pi^\star_{e,t'}$$

Lemma 3 (threshold structure inside a segment, no limit binding)

$$u_t = \alpha_e \pi^\star_{e,t} - \epsilon_e, \quad c_t = \zeta_e + \beta_e \pi^\star_{e,t}: \qquad p^\star_{e,t} = \begin{cases} \bar{p}_e, & \lambda_t < u_t \\ \in [0, \bar{p}_e], & \lambda_t = u_t \\ 0, & \lambda_t > u_t \end{cases} \qquad g^\star_{e,t} = \begin{cases} \bar{g}_e, & \lambda_t > c_t \\ \in [0, \bar{g}_e], & \lambda_t = c_t \\ 0, & \lambda_t < c_t \end{cases}$$

$$\max_{t \in \mathcal{C}_k} \gamma_e^{t-\tau} \frac{\lambda_t + \epsilon_e}{\alpha_e} \le \pi^\star_{e,\tau} \le \min_{t \in \mathcal{G}_k} \gamma_e^{t-\tau} \frac{\lambda_t - \zeta_e}{\beta_e}$$

$\tau \in \mathcal{S}_{e,k}$: any reference interval of the segment (e.g. its marginal interval); $\mathcal{S}^\circ_{e,k} = \mathcal{S}_{e,k} \setminus \mathcal{H}_e$ (no SOC limit binds)

$\mathcal{C}_k = \{t \in \mathcal{S}^\circ_{e,k}: \ p^\star_{e,t} > 0\}$ (charging intervals), $\mathcal{G}_k = \{t \in \mathcal{S}^\circ_{e,k}: \ g^\star_{e,t} > 0\}$ (discharging intervals)

Equality on the left if some charging interval is partial; on the right if some discharging interval is partial.

So which price?

- The most expensive interval in which the ESS still charges (γ = 1), wherever it lies in the segment.

Proof $\nu = 0$ in $\mathcal{S}^\circ_{e,k}$: $u_t - \lambda_t = \bar{\rho}_t - \underline{\rho}_t, \quad \lambda_t - c_t = \bar{\sigma}_t - \underline{\sigma}_t, \quad \bar{\rho}, \underline{\rho}, \bar{\sigma}, \underline{\sigma} \ge 0 \ + \ (\text{CS4})$

Key idea Inside a segment the ESS is a price-taker with two thresholds: buy below u, sell above c, idle in between.

Example: the highest charging price sets the value

$\alpha = 0.9$, $\beta = 1$, $\gamma = 1$, $\varepsilon = \zeta = 1$, $\Delta T = 1$ h, $\underline{p} = \bar{g} = 100$ MW, $\underline{s} = 200$ MWh, $s_0 = 0$, $w = 0$. Prices (0, 10, 5, 50, 50).

interval	1	2	3	4	5
price λ	0	**10**	5	50	50
charge (MW)	100 = $\underline{p}$	**22.2 (partial)**	100 = $\underline{p}$	0	0
discharge (MW)	0	0	0	100 = $\bar{g}$	100 = $\bar{g}$
SOC	90	110	200 = full	100	0
(λ+ε)/α	1.11	**12.22**	6.67	–	–
π (LP)	**12.22**	**12.22**	**[12.22, 49]**	**[12.22, 49]**	**0**
$\underline{\rho} = u - \lambda$	10	0	5	–	–

- Threshold $u = 0.9 \cdot 12.22 - 1 = 10$: intervals priced 0 and 5 are below it ⇒ full power, with power rents $\underline{\rho} = 10$ and 5.
- The partial interval (price 10) sets the value, although it is not the last one. Reordering to (0, 5, 10) gives the same value.
- Two partial charges at different prices in one segment are impossible: both would have to equal the same threshold.



Key idea Using any other price would give a negative power multiplier: the bid would ask the ESS to charge where its bid is below the price.

Before Proposition 4: the pieces it combines

Formulas we use (with their labels)

Segments	$\mathcal{S}_{e,k} = \{h_{e,k-1}, \dots, h_{e,k} - 1\}, \quad \mathcal{S}_{e,N_e+1} = \{h_{e,N_e}, \dots, T\}$	between limits
Lemma 1	$\pi_{e,\tilde{t}} = v_{e,k}(\lambda) = \begin{cases} (\lambda_{n(e),\tilde{t}} - \zeta_e)/\beta_e & \text{(D)} \\ (\lambda_{n(e),\tilde{t}} + \epsilon_e)/\alpha_e & \text{(C)} \end{cases}$	price → value
Lemma 2	$t < t' \in \mathcal{S}_{e,k} \;\Rightarrow\; \pi_{e,t} = \gamma_e^{t'-t}\, \pi_{e,t'}$	carry the value
Lemma 3	the highest charging price in the segment sets v	which price
(S8)	$\pi_{e,T} = w_e\big(s_{e,T} - \bar{\bar{s}}_{e,T}\big) \;\; (= w_e)$	last segment
(14)	$c^g_{e,t} = \zeta_e + \beta_e \max\{\pi_{e,t}, \pi_{e,t-1}/\gamma_e\}, \quad u^p_{e,t} = \alpha_e \min\{\cdot\} - \epsilon_e$	for Cor. 4

Notation

$\mathcal{H}_e$	binding intervals $h_{e,1} < \cdots < h_{e,N_e}$ (an SOC limit binds)
$\mathcal{S}_{e,k}$	segment k; $\mathcal{S}_{e,N_e+1}$ is the last one
$\tilde{t}_{e,k}$	marginal interval of segment k: partial trade, no limit binds
(D) / (C)	discharge- / charge-marginal
$v_{e,k}(\lambda)$	value of segment k, read from the price at $\tilde{t}_{e,k}$
$\tau(t)$	marginal interval of the segment containing t (Cor. 4)
$\gamma_e^{t'-t}$	retention between t and t' ($= 1$ if $\gamma_e = 1$)

Used for Proposition 4 = Lemma 1 (one value per segment) + Lemma 2 (carry it) + (S8) (last segment). Corollary 4 then plugs it into (14).

Proposition 4: value of stored energy from prices

$$\pi^{\star}_{e,t} = \begin{cases} \gamma_e^{\tilde{t}_{e,k}-t}\, v_{e,k}(\lambda^{\star}), & t \in \mathcal{S}_{e,k},\ k \le N_e \\ \gamma_e^{T-t}\, w_e(s^{\star}_{e,T} - \bar{\bar{s}}_{e,T}), & t \in \mathcal{S}_{e,N_e+1} \end{cases} \qquad \text{(Prop. 4)}$$

Assumption 1, and every segment k ≤ N contains a marginal interval. Holds for any optimal multipliers, with that solution's binding set.

Proof

- Step 1 (one value per segment): Lemma 1 gives π at the marginal interval = v(λ★).
- Step 2 (carry it): Lemma 2 with t' = t̃ gives π = γ^(t̃−t)·v for every interval of the segment.
- Step 3 (last segment): (S8) gives π_T = w; Lemma 2 with t' = T.

Remarks

- Several marginal intervals in one segment must agree up to γ: their prices are tied together.
- A segment without a marginal interval: prices only bound π (power-limited or idle intervals).
- The last segment needs none: its value comes from the terminal bid.
- t = 2 and t = 4 in the base case trade partially but are not marginal: an SOC limit stops them.

Key idea One price per segment plus the terminal bid determine π everywhere.

Corollary 4: equivalent bids as functions of prices

τ(t): marginal interval of the segment containing t; last segment: τ := T, v_T := w; t = 1: τ(0) := τ(1).

$$c^{g\star}_{e,t} = \zeta_e + \beta_e \max\left\{\gamma_e^{\tau(t)-t} v_{e,\tau(t)},\ \gamma_e^{\tau(t-1)-t} v_{e,\tau(t-1)}\right\}$$

$$u^{p\star}_{e,t} = \alpha_e \min\left\{\gamma_e^{\tau(t)-t} v_{e,\tau(t)},\ \gamma_e^{\tau(t-1)-t} v_{e,\tau(t-1)}\right\} - \epsilon_e$$

Reading

- Same segment for t−1 and t: max = min, narrow band.
- New segment (binding interval): the band widens.

Proof: Proposition 4 for t and t−1, then Corollary 3

$$\pi^{\star}_{e,t} = \gamma_e^{\tau(t)-t} v_{e,\tau(t)} \ \text{(end of } t\text{)}, \qquad \frac{\pi^{\star}_{e,t-1}}{\gamma_e} = \frac{\gamma_e^{\tau(t-1)-(t-1)} v_{e,\tau(t-1)}}{\gamma_e} = \gamma_e^{\tau(t-1)-t} v_{e,\tau(t-1)} \ \text{(start of } t\text{)}$$

Key idea The bids are now written with prices only: v comes from λ★ at marginal intervals, w from the terminal bid.

Baldick's base case, step by step

6 · BIDS FROM PRICES

	t = 1	t = 2	t = 3	t = 4	t = 5	t = 6
price λ★ ($/MWh)	43.1	50	0	0	50	50
ESS charge / discharge (MW)	p = 100	g = 90	p = 200	p = 22.2	g = 175	g = 25
SOC at end of t (MWh)	360	0	720	800	100	0
$\underline{v}$ / $\bar{v}$ / rent r	0 / 0 / 0	47.89 / 0 / 47.89	0 / 0 / 0	0 / 47.89 / −47.89	0 / 0 / 0	9 / 0 / 9
segment; marginal?	S_1; (C)	S_2; no	S_2; (C)	S_3; no	S_3; (D)	S_4; last
price-based value	(43.1+1)/0.9	–	(0+1)/0.9	–	(50−1)/1	w
π(t) (end of t)	**49**	**1.11**	**1.11**	**49**	**49**	**40**
π(t−1)/γ (start of t)	49	49	1.11	1.11	49	49
offer c = 1 + max	**50**	**50**	**2.11**	**50**	**50**	**50**
bid u = 0.9·min − 1	**43.1**	**0**	**0**	**0**	**43.1**	**35**
λ★ in the band	= u	= c	= u	= u	= c	= c

Data: one bus, T = 6, ΔT = 4 h; renewable at 0 $/MWh; thermal 100 MW at 50 $/MWh; α = 0.9, β = 1, γ = 1, ε = ζ = 1, $\bar{s}$ = 800 MWh, s_0 = 0, $\bar{p}$ = $\bar{g}$ = 1000 MW, w = 40.

Key idea Every λ★ sits on an edge of the band: the ESS sets the price in every interval, and the bids reproduce the solver's multipliers.

Before Section 7: why prices can leave π open

7 · NON-UNIQUENESS · RECALL

Formulas we use (with their labels)

(12)	$r_{e,h} = \pi_{e,h-1}/\gamma_e - \pi_{e,h} \neq 0$ only at $h \in \mathcal{H}_e$	jump at a limit
(13)	empty at $h \Rightarrow r_{e,h} \geq 0$; full at $h \Rightarrow r_{e,h} \leq 0$	sign of the jump
(S5)+(CS4)	$p_{e,t} = \bar{p}_e \Rightarrow u_t \geq \lambda_t$; $p_{e,t} = 0 \Rightarrow u_t \leq \lambda_t$	bounds, not equalities
(S6)+(CS4)	$g_{e,t} = \bar{g}_e \Rightarrow c_t \leq \lambda_t$; $g_{e,t} = 0 \Rightarrow c_t \geq \lambda_t$	
Prop. 4	$\pi_{e,t} = \gamma_e^{\tilde{t}-t} v_{e,k}(\lambda)$ if segment k has a marginal interval	unique case

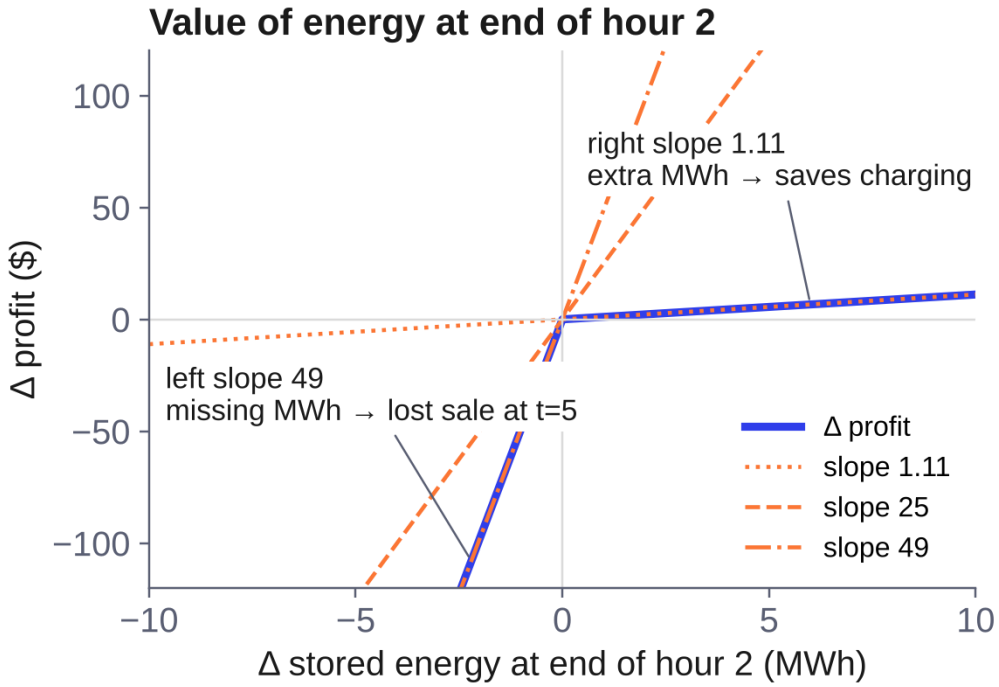
Notation

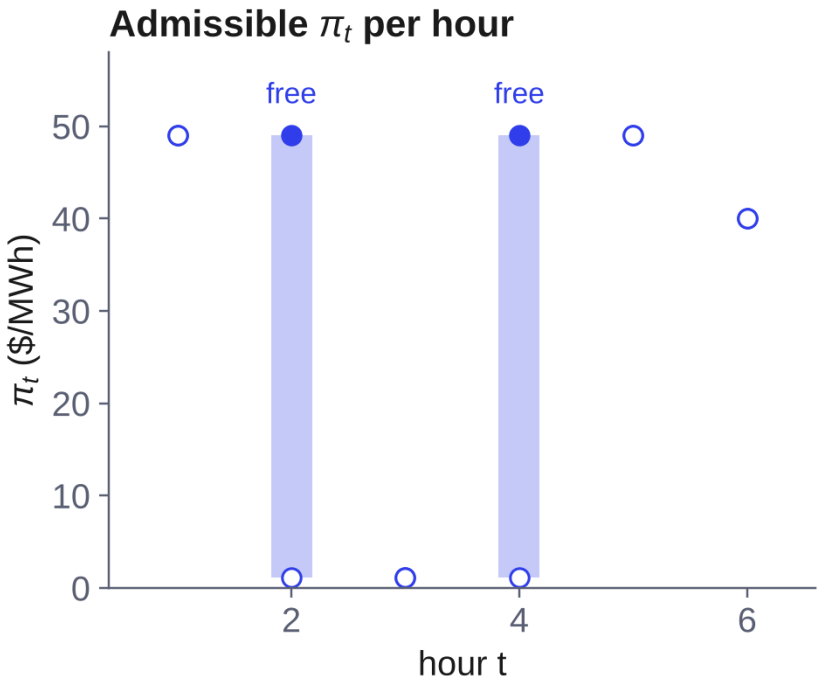
$h \in \mathcal{H}_e$	binding interval (base case: $h = 2$ empty, $h = 4$ full)
u_t, c_t	bid and offer in t (functions of π)
$\pi^-_{e,t}$, $\pi^+_{e,t}$	**new:** lowest / highest admissible value of $\pi_{e,t}$
$\pi^R_{e,t}$	**new:** value chosen by the pre-limit rule
$\mathrm{Proj}_{[a,b]}(x)$	**new:** $\min\{\max\{x, a\}, b\}$: closest point of $[a, b]$ to x

Used for At a binding interval the price only bounds π: the admissible range [π⁻, π⁺] comes from these inequalities.

Where an SOC limit binds, prices give a range for π

7 · NON-UNIQUENESS





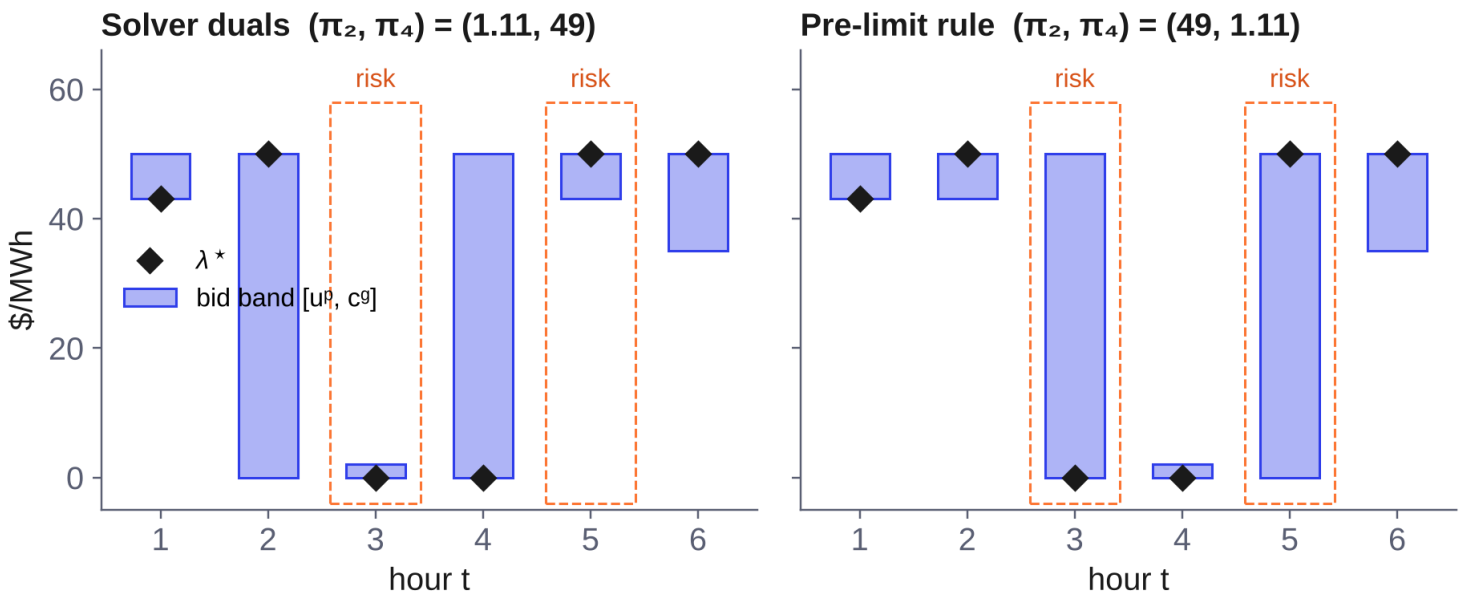
$$\pi_2 \in \Big[\underbrace{\frac{\epsilon}{\alpha} = 1.11}_{\text{extra MWh: saves charging}} \;,\; \underbrace{\frac{\lambda_5 - \zeta}{\beta} = 49}_{\text{missing MWh: lost sale at } t=5} \Big]$$

- The admissible π_t are the right and left derivatives of the optimal value w.r.t. extra stored energy at t.
- Checked by re-solving the LP with ±0.5 MWh.

Key idea The value function has a kink at the limit. Every slope between 1.11 and 49 is a valid π and reproduces the same prices.

Choosing within the range: the pre-limit rule

$$\pi^R_{e,t} = \mathrm{Proj}_{[\pi^-_{e,t},\,\pi^+_{e,t}]}\left(\frac{\pi^R_{e,t-1}}{\gamma_e}\right) \qquad \text{(keep the pre-limit value whenever admissible)}$$



t	planned → unwanted	solver	pre-limit
2	sell → buy	↓ 50	↓ 6.9
3 ⚠	buy → sell while empty	↑ 2.1	↑ 50
4	buy → sell	↑ 50	↑ 2.1
5 ⚠	sell → buy while full	↓ 6.9	↓ 50

Price change from λ★ that triggers the unwanted action.
⚠ = physically impossible (infeasible schedule).

Key idea With perfect forecasts every choice works; with errors, the rule places the wide margins where a mistake would be infeasible.

Before Section 8: the objects the formal results use

Formulas we use (with their labels)

(A8)–(A12)	SOC dynamics, SOC limits, power limits of ESS e	the ESS's constraints
(S5)–(S8)	stationarity of $p,\ g,\ s$ (charge, discharge, value over time, terminal)	dual feasibility
(CS4)–(CS5)	complementary slackness of power and SOC limits	dual $\leftrightarrow$ primal
(E)	$u_t = \alpha_e(\pi_t - \bar{\nu}_t) - \epsilon_e, \quad c_t = \zeta_e + \beta_e(\pi_t + \underline{\nu}_t)$	bid map
Prop. 1	bids (E) from $z^\star \Rightarrow$ SOC dispatch is CD-optimal at $\lambda^\star$	equivalence
Prop. 4	π from prices at marginal intervals	prices $\rightarrow \pi$

Notation

$\mathcal{P}^{\mathrm{SOC}}$, $\mathcal{P}^{\mathrm{CD}}$	market clearing: SOC / current design
$z^\star;\ x^\star$	primal–dual optimum of $\mathcal{P}^{\mathrm{SOC}}$; its dispatch
$\lambda^\star_{n(e)}$	price path at the ESS's bus, $(\lambda^\star_{n(e),1}, \ldots, \lambda^\star_{n(e),T})$
m_e	ESS multipliers $(\pi, \bar{\nu}, \underline{\nu}, \bar{\rho}, \underline{\rho}, \bar{\sigma}, \underline{\sigma})$
θ_e	**new:** technical data submitted in the SOC design
$\mathcal{P}^{\mathrm{SS}}_e(\lambda)$	**new:** the ESS's own price-taking self-schedule
$\mathcal{D}_e(\lambda),\ \Phi_e$	**new:** its optimal multipliers; the set of equivalent bids

Used for Theorem 1 and Proposition 5 apply LP duality (F1)–(F2) to the ESS's own problem, built from exactly these rows.

The ESS's own price-taking problem

Technical data submitted in the SOC design

$$\theta_e = \left(\alpha_e, \beta_e, \gamma_e, \epsilon_e, \zeta_e, \underline{s}_e, \bar{s}_e, s_{e,0}, \bar{p}_e, \bar{g}_e, b_e\right)$$

Price-taking self-schedule at a price path λ

$$(\mathcal{P}_e^{\mathrm{SS}}(\lambda)): \ \min_{p,g,s} \sum_t \Delta T\left[(\epsilon_e + \lambda_t)p_{e,t} + (\zeta_e - \lambda_t)g_{e,t}\right] - b_e(s_{e,T} - \bar{\bar{s}}_{e,T}) \quad \text{s.t. (A8)–(A12) for } e$$

Optimal multipliers D_e(λ), bid map E and set of equivalent bids Φ_e

$$E(m) = (u, c): \ u_t = \alpha_e(\pi_t - \bar{\nu}_t) - \epsilon_e, \ c_t = \zeta_e + \beta_e(\pi_t + \underline{\nu}_t); \qquad \Phi_e(\theta_e, \lambda) = \{E(m): \ m \in \mathcal{D}_e(\lambda)\}$$

Two LP facts used throughout (Boyd & Vandenberghe, §5.5)

- (F1) Every primal optimum and every dual optimum satisfy the KKT conditions together.
- (F2) For a fixed primal optimum $\hat{x}$, the dual optimal set = dual-feasible points satisfying complementary slackness with $\hat{x}$.

Key idea Assumption 2: b_e concave piecewise linear, $\bar{p}, \bar{g} > 0$, P_SS feasible and bounded, so it is an LP with strong duality.

Theorem 1: bids from technical data and prices

Lemma 4 (decomposition)

If z★ is optimal for P_SOC, the schedule of ESS e is optimal for P_SS(λ★ at node n(e)), and its SOC-design multipliers belong to D_e(λ★ at node n(e)). The ESS variables appear only in (A1) at n(e) and in (A8)–(A12).

Theorem 1

For each ESS, take any optimal multiplier vector m_e of its self-schedule at λ★ (any element of D_e), and submit the bids E(m_e) with block sizes equal to the power limits. Then:

- (i) the SOC-design dispatch x★ is optimal for P_CD, and λ★ is an optimal vector of CD prices;
- (ii) the bids depend on the instance only through θ_e and λ★ at node n(e): not on other participants, the network, the ramps, or the ISO's solver;
- (iii) bids (E) of Proposition 1 are the selection m_e = SOC-design multipliers.

Key idea Any optimal dual of the ESS's own problem works: the ESS does not need to know Baldick's multipliers, only the prices.

Theorem 1: proof and what it does not claim

1. Lemma 4: the SOC-design schedule is optimal for P_SS(λ★).
2. (F1): that schedule and the chosen m_e satisfy (S5)–(S8), (CS4)–(CS5) together.
3. Substitute E(m_e) into (S5)–(S6): this is (S6-CD)–(S5-CD); $\hat{p} = \bar{p}$, $\hat{g} = \bar{g}$ make (CS4) = (CS4-CD).
4. Rest-of-market rows contain no ESS multiplier: they hold with z★'s multipliers.
5. The full CD KKT system holds at (x★, λ★); P_CD is an LP ⇒ optimal.

Not claimed

- The ESS's own optimal schedule may differ (degenerate LP); bids are unaffected (F2).
- λ★ is a CD clearing price; if the CD dual is not unique, the ISO's pricing rule could pick another one (not the case in Baldick's examples).
- x★ is a CD optimum; quantities may need a tie-break rule.



Key idea The proof only uses LP duality: no assumption on the rest of the market beyond convexity.

Proposition 5: all equivalent bids, from prices and one schedule

Fix λ and any optimal schedule $(\hat{p}, \hat{g}, \hat{s})$ of P_SS(λ). Tight SOC rows:

$$\mathcal{U} = \{t : \gamma_e \hat{s}_{t-1} + \Delta T \alpha_e \hat{p}_t = \bar{s}_e\}, \qquad \mathcal{L} = \{t : \gamma_e \hat{s}_{t-1} - \Delta T \beta_e \hat{g}_t = \underline{s}_e\}$$

(u, c) belongs to Φ_e(θ_e, λ) if and only if $u = \alpha(\pi - \bar{v}) - \varepsilon$, $c = \zeta + \beta(\pi + \underline{v})$ for some π, $\bar{v}$, $\underline{v}$ with:

$$\begin{aligned}
&\text{(a)} \quad \pi_T \in \partial b_e(\hat{s}_T - \bar{\bar{s}}_{e,T}) \quad (\pi_T = w_e) \\
&\text{(b)} \quad \pi_t = \gamma_e(\pi_{t+1} - \bar{\nu}_{t+1} + \underline{\nu}_{t+1}), \ t < T \\
&\text{(c)} \quad \bar{\nu}_t = 0 \text{ for } t \notin \mathcal{U}, \quad \underline{\nu}_t = 0 \text{ for } t \notin \mathcal{L}, \quad \bar{\nu}, \underline{\nu} \geq 0 \\
&\text{(d)} \quad u_t \ \{\leq, =, \geq\} \ \lambda_t \ \text{ if } \ \hat{p}_t \{= 0, \in (0, \bar{p}), = \bar{p}\}; \quad c_t \ \{\geq, =, \leq\} \ \lambda_t \ \text{ if } \ \hat{g}_t \{= 0, \in (0, \bar{g}), = \bar{g}\}
\end{aligned}$$

Proof

- (F2): dual optimal ⇔ dual feasible + CS with $(\hat{p}, \hat{g}, \hat{s})$.
- (a) = (S8), (b) = (S7), (c) = (CS5).
- (d): eliminate ρ, σ from (S5)–(S6) with (CS4); conversely set $\bar{\rho} = [u - \lambda]^+$ etc.

Key idea Φ_e is a polyhedron computed from θ_e and λ alone (one LP, then linear conditions), and it is the same for every optimal schedule.

Corollary 5: when prices pin π, using the schedule only

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$$\text{(i)} \quad t \notin \mathcal{U} \cup \mathcal{L} \;\Rightarrow\; \pi_{t-1} = \gamma_e \pi_t$$

$$\text{(ii)} \quad 0 < \hat{p}_t < \bar{p}_e,\; t \notin \mathcal{U} \;\Rightarrow\; \pi_t = \frac{\lambda_t + \epsilon_e}{\alpha_e}; \qquad 0 < \hat{g}_t < \bar{g}_e,\; t \notin \mathcal{L} \;\Rightarrow\; \pi_t = \frac{\lambda_t - \zeta_e}{\beta_e}$$

$$\text{(iii)} \quad \pi_T = w_e \;(\text{linear } b_e)$$

Chains of intervals linked by (i) that contain a type-(ii) interval or T have a unique π; other chains are only bounded, with r ≥ 0 where only the lower row is tight and r ≤ 0 where only the upper row is tight.

Base case, from θ and λ★ only	t = 1	t = 2	t = 3	t = 4	t = 5	t = 6
tight row	L	L	L	U	U	L
type	(ii) charge	–	(ii) charge	–	(ii) discharge	(iii)
π_t	**49**	**[1.11, 49]**	**1.11**	**[1.11, 49]**	**49**	**40**

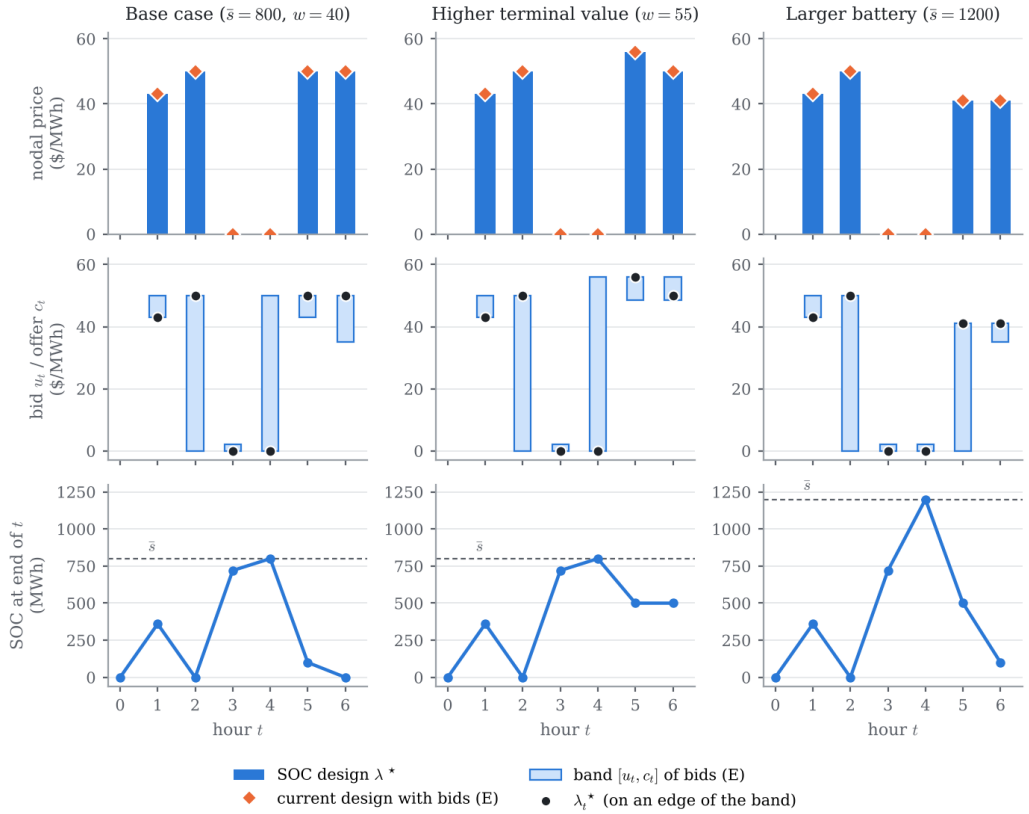
- t = 2: $r_2 = \pi_1 - \pi_2 \geq 0$ and $r_3 = \pi_2 - \pi_3 \geq 0 \Rightarrow 1.11 \leq \pi_2 \leq 49$.
- t = 4: $r_4 = \pi_3 - \pi_4 \leq 0$ and $r_5 = \pi_4 - \pi_5 \leq 0 \Rightarrow 1.11 \leq \pi_4 \leq 49$.
- Same ranges as the dual optimal face and the perturbation analysis.

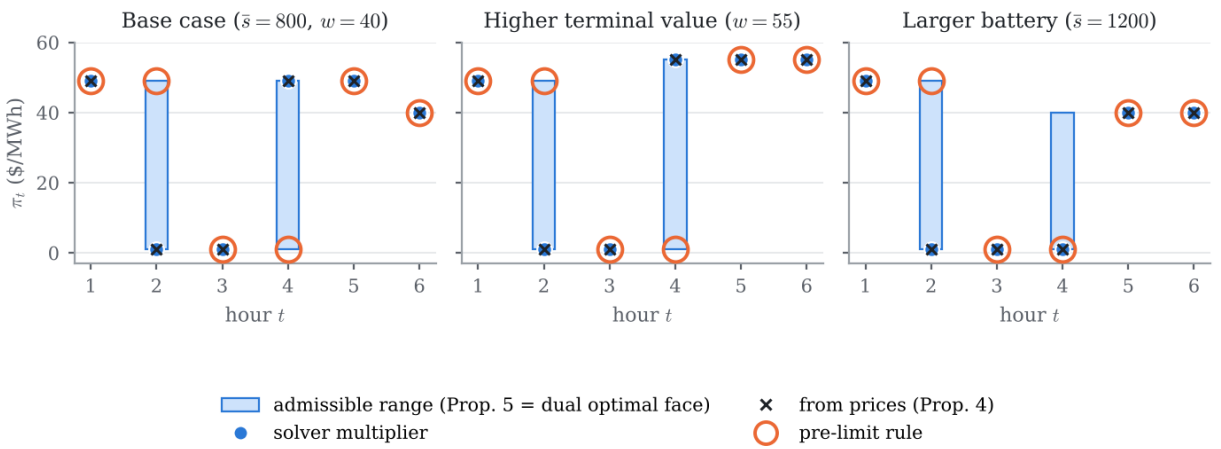
Key idea Corollary 5 is Proposition 4 without any dual information: the tight sets come from the schedule, which anyone can compute.

Numerical verification

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Baldick's three single-bus cases: base ($\bar{s}$ = 800, w = 40), higher terminal value (w = 55), larger battery ($\bar{s}$ = 1200).





- **Same prices in both designs** largest price difference ≤ $3 \cdot 10^{-15}$ $/MWh (Prop. 1)
- **Baldick's dispatch is CD-optimal** objective gap 0; f_SOC = f_CD + K (Props 1–2)
- **π rebuilt from prices** Prop. 4 = solver multipliers exactly
- **Ranges from prices + one schedule** Prop. 5 = dual optimal face (≤ $7 \cdot 10^{-15}$); Thm. 1 at every extreme point

Key idea λ★ sits on an edge of every bid band: the SOC schedule lies on the bid curves. Script: numerical_check.py (scipy / HiGHS).

What the results are used for

FROM THEORY TO USE

$$\Delta SW = SW^{\mathrm{SOC}} - SW^{\mathrm{CD}}(\mathcal{B}) \geq 0, \qquad \mathcal{B} \in \{\mathcal{B}^{\star},\ \mathcal{B}^{\mathrm{priv}},\ \mathcal{B}^{\mathrm{fcst}} = \Phi_e(\theta_e, \hat{\lambda})\}$$

B★ exact bids

Baldick's multipliers: sanity check, ΔSW = 0.

B-priv private bids

v = 0, π = γ^{T−t}w: no forecast; exact only if no limit binds.

B-fcst forecast bids

Φ(θ, λ̂) with the pre-limit rule: realistic bidder.

For regulators

- ΔSW with forecast bids = the value of adopting the SOC design.
- Baldick's model is run ex post as the benchmark.

For market operators

- Φ(θ, ISO forecast) = an SOC-consistent reference bid for monitoring and mitigation.
- The admissible range is a natural tolerance band.

Key idea One map, Φ(technical data, prices), serves the battery owner, the regulator and the market monitor.

SUMMARY

The results in one page

1. Prop. 1–3: bids built from the SOC design's multipliers reproduce its dispatch and prices in the current design.

2. Cor. 3: the bids only need π at the start and end of each interval: $c = \zeta + \beta \cdot \max$, $u = \alpha \cdot \min - \varepsilon$.

3. Lemmas 1–3, Prop. 4, Cor. 4: π is the price of the marginal interval of its segment (the highest charging / lowest discharging price), carried by γ.

4. Non-uniqueness: where an SOC limit binds, prices give a range; the pre-limit rule selects within it.

5. Thm. 1, Prop. 5: no multipliers needed; all equivalent bids = Φ(technical data, local prices), a polyhedron.

Reference: R. Baldick, "A state-of-charge based formulation for storage participation in electricity markets," arXiv:2606.14993, 2026.


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